

Exam 4: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (18 pts) A ZIRLO cladding tube is in reactor at 625 K for 500 days. The initial wall thickness is 600 μm .
 - a) Estimate the oxide thickness after this time?
 - b) Assuming the hydrogen pickup fraction is 14%, what is the weight ppm of hydrogen in the cladding? Assume $\text{PBR} = 1.56$, $\rho_{\text{Zr}} = 6.5 \text{ g/cc}$, $\rho_{\text{ZrO}_2} = 5.68 \text{ g/cc}$.
2. (2 pts) What is the rate-limiting step in the aqueous corrosion of Zr cladding?
3. (8 pts) Where do hydrides form in the cladding? Why? What impacts do hydrides have?
4. (12 pts) What is a RIA? What is a typical RIA in a PWR or BWR. Describe what happens during a RIA.
5. (12 pts) What is a LOCA? Describe the impacts of a LOCA. How is it different than a RIA?
6. (4 pts) How does burnup impact the type of failure and the failure probability during an accident?
7. (6 pts) What are the four pathways to make the fuel/cladding system more accident tolerant?
8. (6 pts) List one near-term ATF concept and why it is of interest.
9. (6 pts) What happens to zirconium cladding when it is exposed to high temperature steam?
10. (6 pts) List three examples of limiting phenomena governing LWR operation.
11. (6 pts) What fuel performance and safety impacts does CRUD have?
12. (6 pts) List two water chemistry controls that have been implemented in LWRs, including why they were implemented.
13. (8 pts) List and describe two key microstructural phenomena that are unique to MOX fuels.

$T = 625K$, $t = 500 \text{ days}$, $t_{\text{half}} = 600 \mu\text{m}$

a.) see if we are beyond transition time:

$t^*(d) = 6.62 \times 10^{-7} e^{11949/625K} = 133 \text{ days} < 500 \text{ days}$

$\delta^* = 5.1 e^{-550/625} = 2.115 \mu\text{m}$

$u_L = 7.49 \times 10^6 e^{-12500/625} = .01542$

$\delta(\mu\text{m}) = 2.115 \mu\text{m} + .01542 \frac{\mu\text{m}}{d} (500 - 133)$

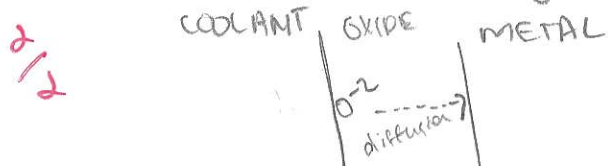
$\delta = 7.77 \mu\text{m}$

b.) $S = .14$, $PBR = 1.56$, $S_{Zr} = 6.5 \frac{g}{cc}$, $S_{ZrO_2} = 5.68 \frac{g}{cc}$

$$C_H^{\text{clad}} = \frac{2(.14)(7.77 \mu\text{m})(5.68 \frac{g}{cc})}{(600 \mu\text{m} - \frac{7.77 \mu\text{m}}{1.56}) 6.5 \frac{g}{cc}} \left(\frac{32}{32+91} \right) \left(\frac{1}{16} \right) \times 10^6$$

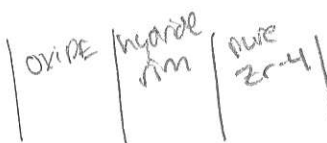
$C_H^{\text{clad}} = 51.93 \text{ wt. ppm}$

2.) diffusion of oxygen through the oxide layer



$\frac{8}{8}$

3.) hydrides typically form in areas of low temperature and high tensile stress, creating a "hydride rim" that exists between the oxide layer and the Zr-4 layer of the cladding. Hydrides can also form blisters. The hydride rim is formed there because it is a lower temperature region with high levels of tensile stress because of axial expansion of the fuel pellet, creating fuel/clad mechanical



3 continued interaction. Hydrogen migrates to low temperature region because precipitation occurs at a lower hydrogen concentration at lower temperatures. Hydrogen migrates to high tensile stress regions because the volume of the crystal lattice is higher there, which allows interstitial hydrogen to exist at a lower energy. Effects of hydrides - are: decreased fracture toughness, hydride embrittlement, delayed hydride cracking, accelerated corrosion and irradiation growth.

4.) RIA is a reactivity insertion accident which serves as a design basis accident for LWR's. In a PWR, a typical RIA occurs when control rod housing fails, causing the control rod to be completely ejected out of the core b/c of coolant pressure (CHCA). In a BWR, an RIA occurs when the control rod drive mechanism fails, causing the control rod to completely drop out of the core (CRDA). Worst case RIA's happen at HZP in a PWR and CZP in a BWR. Sequence of events of an RIA:

- 1.) rapid reactivity insertion that is quickly brought back down w/ negative feedback mechanisms ← Doppler/moderator density feedback
 - 2.) rapid rise in fuel temp, leading to rapid solid pellet thermal expansion and rapid fission product swelling
 - 3.) this can cause fuel failures via various mechanisms
 - PCM (typically occurs for higher burnup rods, occurs because clad stress is increased quickly from expansion)
 - clad ballooning or burst (typically occurs at lower burnups as a result of DNB, when clad temp increases greatly)
- both types of fuel failure exacerbated by cladding oxidation/hydride state

- worst case RIA is when hot fuel ejected into coolant, causing immediate pressure shock to core

3.) Loss of coolant Accident, another DBA for LWR. ¹²/₁₂

sequence of events: Coolant is lost via a pipe break or something similar. Immediately, pressure in the core drops. Reactor immediately scrams and emergency cooling activated. No more fission heat, but decay heat is still produced. Fuel rods experience ballooning and burst failure because internal pressure is higher than external pressure and high temperature steam oxidation causes significant embrittlement. This can also block coolant flow, creating an uncoolable geometry. LOCA is different than a RIA because it occurs on a much longer timescale. (RIA occurs in milliseconds). LOCA's also have the concern of high temperature steam oxidation and breakeaway reactions. ⁴/₄

6.) Kind of already explained this in question 4, but with increasing burnup, we have increased embrittlement and loss of fracture toughness in the cladding because of oxidation and hydriding. At increasing burnup we also have pellet clad contact, which makes PCC failure a greater probability at higher burnups. At lower burnups, clad ballooning and burst is a more likely failure event. Overall, failure probability increases with burnup because rods fail at lower peak enthalpy and stress levels. (consider that in the case of an RIA, lower burnup rods have higher peak enthalpies, but this is counteracted by clad embrittlement/other cladding burnup effects.)

7.) ⁴/₆

- 1.) improved fuel properties (lower fuel temps/higher conductivity, higher melting temp)
- 2.) improved cladding properties (higher fracture toughness, less PCC/PCCI)
- 3.) enhanced fission product retention
- 4.) improved steam oxidation kinetics (lower rate, lower heat rate)

8.) Metallic coatings of Zr cladding. Metallic coating of interest contains Cr. Coating applied via cold spray or other mechanisms. Issues are metal coating adherence to cladding and Zr-Cr eutectic reaction at high temps. This concept is of interest because it significantly improves oxidation rates, especially at high temperature.

9.) at high temperatures, oxidation rates greatly increase b/c rxn is exothermic. If heat cannot be carried away as quickly as it is produced, runaway reaction can occur, leading to a completely oxidized cladding. Also, at high temperatures, Zr transforms into Beta phase w/ an oxygen stabilized alpha phase at the oxide/metal interface, causing further embrittlement and potentially superplastic behavior.

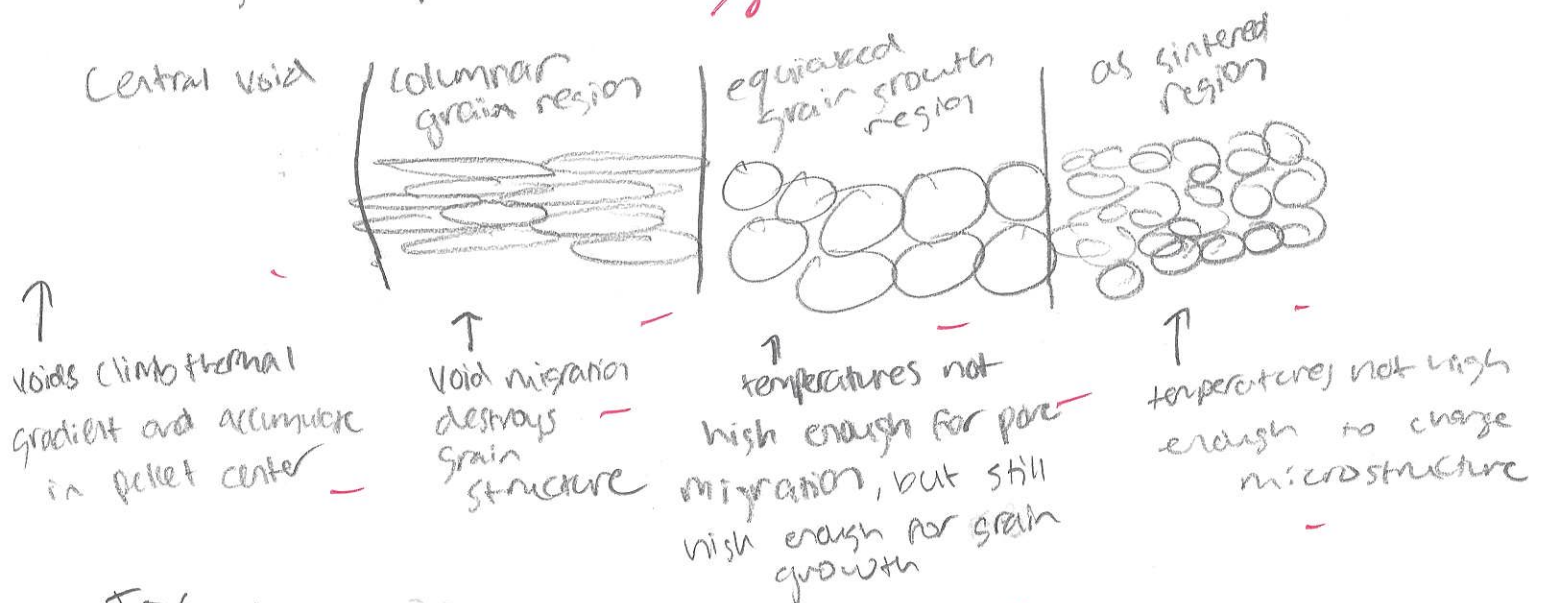
- 10.)
- 1.) PCMI: tensile stress levels are limited
 - 2.) DNB: departure from nucleate boiling ratio monitored, creates limits on thermal power
 - 3.) cladding oxidation and hydriding: levels of oxidation are limited (less than 10% of Zr cladding can be consumed)

12.) 1.) Zinc injections used in both PWR and BWR's for radiation control. Zinc is more likely to incorporate into the spinel structure on core internals than crud constituents (specifically, CoO), resulting in less CoO on reactor internals that can be activated. Also zinc forms protective layer on core internals, preventing preexisting radioactive CRUD from flaking off + releasing.

2.) pH control in PWR's with addition of LiOH . Alkaline solutions (keeping pH above 6.9) leads to less corrosion and CRUD buildup. LiOH added to counteract the pH decrease b/c of boric acid used for reactivity control.

11.) CRUD decreases the thermal conductivity of the fuel rods, causing higher fuel temperatures. CRUD can also lead to CRUD induced power ^{shifts} trips in PWR's by trapping boron when CRUD constituents become activated because of close proximity to neutron flux, they become radioactive, which creates radiation dosage safety issues when CRUD flakes off core internals and escapes into reactor water and builds up on core externals. $\frac{6}{6}$

13.) Restructuring: 4 regions form in the fuel pellets b/c of high temperatures $\frac{8}{8}$



JOG (Joint Oxide Growth Formation): even after gap closure, tiny gap spaces still exist b/c of surface roughness. Solid fission products migrate towards this area, forming a layer of oxide between the clad and fuel, which helps decrease PWR and stress, leads to less fission product swelling, but can create PCCI issues.

